

This issue

Brings you awareness about department activities done by staff and students.



ABOUT THE INSTITUTE

Sagi Rama Krishnam Raju Engineering College, established in 1980, is a self-financed academic institution of coeducation striving to provide a high-quality technical education to engineering aspirants. Being one of the premier and well-established technical institutions in the country, it continues to render service to the nation and the world at large with its alumni holding highly prestigious positions and making a substantial contribution. It is an autonomous Institution (UGC Approved) and permanently Affiliated with Jawaharlal Nehru Technological University Kakinada (JNTUK). The college is replete with lush coconut trees spanning across a plot area of 30 acres at China Amiram, Bhimavaram, on the Bhimavaram-Vijayawada state highway and it is also well-connected to all other places by road and rail transport. Over the years, the college has established itself as a premier institution imparting technical education of a very high standard leading to the B. Tech. degrees in 11 branches of engineering, P.G degrees in 6 specializations, and 7 Ph. D. programs. S.R.K.R. Engineering College was established with a mission to develop the institution into a world-class Centre of excellence in the field of technical education that fosters research and essentials of employability and strives for pre-eminence amongst the premier institutions with high commitment to the socio-ecological well-being of the nation.

RECOGNIZED AS NPTEL DISCIPLINE STAR

Dr. M. Satish Kumar, Assistant Professor in the Department of Electronics and Communication Engineering (ECE), was specially recognized as a **Discipline Star by NPTEL** for motivating and actively participating in certification courses within the same discipline. He received a Certificate of Appreciation for being recognized as an **"NPTEL Discipline Star"** in **Electrical Engineering from NPTEL (a project of MHRD initiated by seven IITs and IISc)** for the period **July to December 2025**.

For this achievement, he successfully completed 56 weeks of NPTEL courses in Electronics and Communication Engineering by completing more than 50 weeks of learning, securing marks greater than 55% in every course offered by NPTEL.



CHIEF EDITOR

Dr. S. S. MOHAN REDDY
HOD, ECE

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Smt. K. L. DIVYA

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M.YUGANDHAR, 3RD ECE
M.SIVA SAI PRAKASH, 3RD ECE
P. JAYANTH, 2ND ECE
Y. HEMANTH, 2ND ECE

AICTE-QIP-PG-CERTIFICATE PROGRAMME IN CYBERSECURITY

The Department of Electronics and Communication Engineering congratulates **Dr. P. Krishna Kanth Varma, Associate Professor**, on successfully completing the **AICTE Quality Improvement Programme (QIP) Postgraduate Certification Programme in Cybersecurity** conducted by **IIIT Vadodara – International Campus Diu, Gujarat**, from **23 June 2025 to 27 December 2025**. The **six-month** programme, attended by **38 faculty members** from across the country, covered important topics such as **Programming for Cybersecurity, Data Science, Information Security, Cybersecurity Essentials, and Cybersecurity Tools and Technologies**. As part of the programme, **Dr. Krishna Kanth Varma** completed a project titled **"Detecting DDoS Attacks in the Software-Defined Networking (SDN) Environments Using Time-Based Features** and demonstrated on **Wi-Fi security assessment using the Aircrack-ng suite**. He completed programme with an outstanding **CPI of 10.0/10.0**, and reflecting his dedication to continuous learning and professional excellence in the field of **cybersecurity**.

DEPARTMENT VISION
Envision a diverse and stimulating academic research ambience for the student community and shaping them into competent professionals in the field of Electronics and Communication Engineering and to cater to the needs of society with a keen sense of environmental consciousness.



AICTE-QIP-PG-CERTIFICATE PROGRAMME IN DEEP LEARNING: FUNDAMENTALS AND APPLICATIONS

The Department of Electronics and Communication Engineering congratulates **B. Tapasvi**, Assistant Professor, on successfully completing the **AICTE Quality Improvement Programme (QIP) Postgraduate Certification Programme in Deep Learning: Fundamentals and Applications** conducted by **Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, Gujarat**, from **16 June 2025 to 23 December 2025**. The six-month programme was attended by **47 faculty members** from across the country and covered key topics including **Essential Mathematics for Deep Learning, Python Programming, Machine Learning, Neural Networks, Deep Learning Architectures, and a Mini Project**. The programme was coordinated by **Dr. Kamal Captain**, with sessions delivered by **Dr. Kamal Captain, Dr. Kishore Upla, Dr. Suman Deb, and Prof. J. N. Sarvaiya**. It was designed to provide faculty members with a strong foundation in **deep learning** and **practical exposure to AI technologies for teaching and research**. The successful completion of this certification reflects **Ms. Tapasvi's** commitment to continuous learning, professional development, and excellence in emerging technologies.

DEPARTMENT MISSION

- To educate the students with the state of the art technologies in Electronics and Communication Engineering to meet the ever growing challenges of the industry.
- To nurture the spirit of innovation and creativity in the faculty and students in order for them to carry out research in collaboration with research organizations and industry
- To provide ethical and value based education that promotes activities pertaining to societal needs.



AICTE QIP PG Certificate Programme			
Period: June, 2025 to December, 2025			
AICTE ID: 1-2657492188		Name: Mr. Bokka Tapasvi	
Course	Credit (T + P)	Grade	Grade Point
Essential Mathematics for Deep Learning and Introduction to Python Programming	3	A	9
Machine Learning	4	A+	10
Neural Network and Deep Learning	4	A	9
Advanced Deep Learning Architectures and Applications	3	A+	10
Mini-Project	4	A	9
Total Credits: 18		Attendance: 100%	CGPA: 9.39
Grade	Grade Points	Grade Percentage Range	
A+	10	90% - 100%	
A	9	80% - 90%	
B	8	70% - 80%	
C	7	60% - 70%	
D	6	50% - 60%	
F	0	Below 50%	
Calculation of CGPA: $\frac{\sum (\text{Course Credit} \times \text{Grade Point})}{\sum (\text{Total Course Credit})}$			



GUEST LECTURE ON LIFE SKILLS, GLOBAL OPPORTUNITIES, AND MORAL VALUES

On January 23, 2026, the ECE Department, in association with the SRKREC Alumni Association, hosted an enriching guest lecture on "Life Skills, Global Opportunities, and Moral Values." The session was led by distinguished alumnus **Mr. Chandra Sekhar Nallam, Founder and CEO of eNcloud Services LLC**. Graced by **HOD Dr. S. S. Mohan Reddy** and retired professors **Dr. K. V. S. N. Raju** and **Dr. D. V. R. Mohan**, the event offered students invaluable insights into holistic personal and professional development. **Mr. Nallam** drew from his global corporate journey to guide students on **higher education prospects abroad, diverse career domains, and industry expectations**, while emphasizing core values like **integrity, leadership, and adaptability**. The highly interactive session concluded with a special felicitation of the guest speaker, leaving **ECE students** inspired to build strong technical skills grounded in **ethical values**.



ABOUT OUR DEPARTMENT

The department of ECE was established in the year 1980 with an intake of 35 which was subsequently raised to 240 over a period of time. M. Tech course in Communication Systems was started in 2006. The department is presently offering 4-year B. Tech program in ECE, 2-year M. Tech in Communication Systems and Ph.D. program. The department has highly qualified, experienced and dedicated staff members. The department has dedicated technical and non-teaching staff members. The department has good infrastructure facilities. The department was accredited four times by National Board of Accreditation (NBA). The department has student chapters of professional bodies like IETE and college level ISTE, which conduct various student technical activities. Students of ECE participate in National level technical paper and hardware competitions being organized at various other colleges and some of them are awarded with prizes. They have been obtaining good scores in GRE, TOEFL and CAT. Students are placed in companies like TCS, Polaris, CTS, IBM, Infosys, Texas Instruments, BSNL, BEL, ISRO and DRDO. ECE department has been identified as research center by JNT University. The research center provides required facilities for carrying out research work for scholars. The department is associated with organizations like DRDO, NSTL, BEL, ISRO, BSNL, NRSA and RTTC. The department has MOUs with DS Reddy Consultancy, ETI Lab Pvt. Ltd., Salcit Technologies, Hyderabad and TCS for collaborative work and faculty improvement programs. Faculty published and presented about 512 research papers in technical journals and conferences.

INNOVATING THE SILENT BACKBONE OF POWER TRANSMISSION

– THE ROLE OF A COMMUNICATION ENGINEER

SRKR Engineering College (A), Bhimavaram, successfully organized an interactive technical session titled **“Innovating the Silent Backbone of Power Transmission – The Role of a Communication Engineer”** on **27 February 2026** at the **ECE Seminar Hall**. The programme was conducted as an **IIC 8.0 MIC Driven Activity** under the **Pre-Summit Engagements of the India AI Impact Summit 2026**. It was organized by the **SRKREC-Institution’s Innovation Council (IIC)** in association with the **Department of Electronics and Communication Engineering** and **SRKREC Alumni Association**, with a focus on **innovation, AI talent development, and applied technology**. The session was convened by **Dr. S. S. Mohan Reddy, Head, ECE**, and coordinated by **Dr. G. Nagaraju, Associate Professor, ECE**. The guest lecture was delivered by **Sri Pilla Mahesh, Deputy Executive Engineer, Telecom, APTRANSCO, Visakhapatnam Circle**. He explained the significant role of **communication engineers in modern power transmission systems** and highlighted how **advanced communication technologies enable real-time monitoring, grid stability, renewable energy integration, and prevention of system failures**. The resource person further discussed the responsibilities of communication engineers in areas such as **System Protection, SCADA, Data Acquisition, and Cybersecurity, emphasizing their role in building resilient and intelligent power infrastructure**. The session provided valuable technical insights and an opportunity for students to interact with an industry expert. A total of **105 second- and third-year B.Tech ECE students** and **25 faculty members** actively participated in the programme. The event concluded with a vote of thanks proposed by **M. Usha, II B.Tech ECE**.



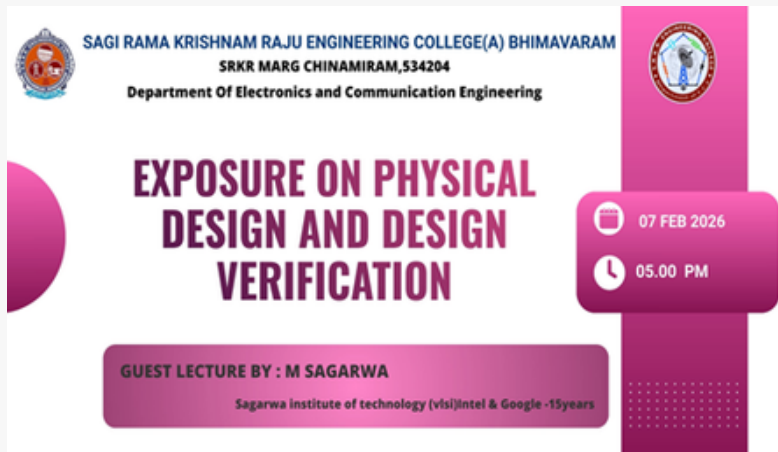
S.R.K.R Engineering College(A),CHINNAMIRAM, BHIMAVARAM
Department of Electronics and Communication Engineering
in Association with
SRKREC Alumni Association
WELCOME
TO
GUEST LECTURE
OF
“INNOVATING THE SILENT BACKBONE OF POWER TRANSMISSION”
– THE ROLE OF A COMMUNICATION ENGINEER



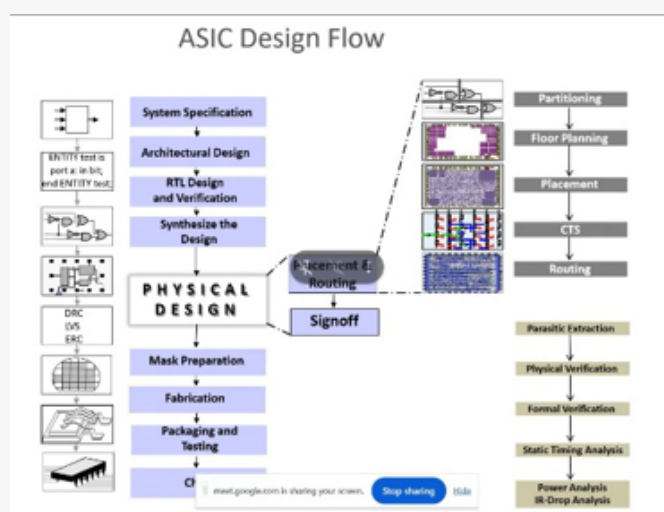
RESOURCE PERSON
Sri. Mahesh Pilla
(1999 - 2002 ECE Batch)
Deputy Executive Engineer, TELECOM,
APTRANSCO,
VISAKHAPATNAM Circle.

Date: 27th, February 2026
Time: 10:30 am to 12:30 pm
Venue: ECE Seminar Hall

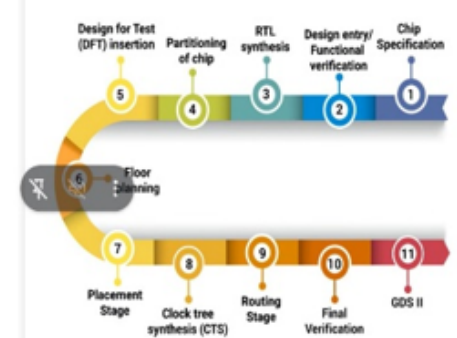
GUEST LECTURE ON “EXPOSURE ON PHYSICAL DESIGN AND DESIGN VERIFICATION”



The Department of Electronics and Communication Engineering, Sagi Rama Krishnam Raju Engineering College (Autonomous), Bhimavaram, organized a guest lecture titled “Exposure on Physical Design and Design Verification” on 07 February 2026 at 5:00 PM through an online platform. The session was coordinated by K. Bala Sindhuri and Y. S. S. Sriramam from the Center of Excellence in VLSI Design. The lecture was delivered by Mrs. M. Sagarwa from Sagarva Institute of Technology, an industry expert with over 15 years of experience in the VLSI domain, including professional experience with leading organizations such as Intel and Google. The session provided students with practical insights into Physical Design and Design Verification, highlighting their importance in semiconductor manufacturing and real-time chip development. The resource person explained the VLSI design flow, industry tools, and the role of verification in ensuring chip accuracy and reliability before fabrication. Around 30 students actively participated and interacted with the expert, gaining valuable guidance on VLSI career opportunities and industry expectations. The session helped bridge the gap between theoretical learning and practical industry applications. The lecture enabled students to develop a basic understanding of Physical Design and Design Verification, gain introductory knowledge of the VLSI design flow, and understand the importance of verification in chip development. It also created interest among students in exploring careers in the VLSI, semiconductor, and electronics sectors. The programme contributed to SDG 4: Quality Education by providing technical exposure and SDG 9: Industry, Innovation and Infrastructure by introducing students to semiconductor technologies and innovation. Overall, the session served as a valuable introductory platform for students to explore the growing opportunities in the VLSI domain.



How does the ASIC design cycle work?



INDUSTRIAL VISIT

SATISH DHAWAN SPACE CENTRE (SDSC SHAR), ISRO



The Department of Electronics and Communication Engineering (ECE), SRKR Engineering College, Bhimavaram, organized an industrial visit to **Satish Dhawan Space Centre (SDSC SHAR), Sriharikota**, on **10 March 2026** to provide students with practical exposure to advanced **space technologies, satellite communication, and launch vehicle operations**. During the visit, students explored the Mission Control Centre, Launch Pad-1, Launch Pad-2, and **ISTRAC Ground Station**, gaining valuable insights into mission monitoring, launch operations, telemetry, satellite tracking, communication systems, and real-time applications of **ECE concepts** in aerospace technology. The visit also included visits to **Pulicat Lake** and **Changalamma Temple**, offering students opportunities for **ecological and cultural learning**. Overall, the visit successfully bridged theoretical knowledge with practical applications, enhanced students' technical awareness, and motivated them to **explore career opportunities** in **space technology, satellite communication, aerospace systems**, and related fields.



FACULTY DEVELOPMENT PROGRAM (ONLINE)

ABOUT SRKR ENGINEERING COLLEGE

Sagi Rama Krishnam Raju Engineering College (SRKREC), established in 1980, is a self-financed academic institution of co-education striving to provide a high-quality technical education to engineering aspirants. Being one of the premier and well-established technical institutions of the country, it continues to render service to the nation and the world at large with its alumni holding highly prestigious positions and making substantial contributions. The president of the institution Sri. S. Prasada Raju carries forward the vision of his father Late Sri Sagi Rama Krishnam Raju, a philanthropist, driving the institution on the academic path of excellence. The college is replete with lush coconut trees spanning across a plot area of 30 acres at Chinna Amiram, Bhimavaram, on the Bhimavaram- Vijayawada state highway and it is also well-connected to all other places by road and rail transport. Over the years, the college has established itself as a premier institution imparting technical education of a very high standard leading to the **B.Tech degrees in 12 branches of Engineering, P.G degrees in 6 specializations, and 10 Ph. D. programs.**

HIGHLIGHTS OF SRKREC

Accredited by NAAC A* Grade
Recognized as SIRO
NBA accreditation for six branches.
Fourteen centers of excellence.
JNTUK recognized research centers for ECE, CE, ME, EEE, CSE, IT, Mathematics, and Chemistry.

ABOUT DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

The Department of Electronics and Communications Engineering was established in the year 1980 with an intake of 35 students and the number was subsequently raised to 240 students over the last four decades. The department is presently offering 4-year B. Tech program in ECE, 2-year M.Tech program in Communication Systems besides the Ph.D. program. It is recognized as a research center by JNTU, Kakinada. The department has student chapters of professional bodies such as IETE. These bodies regularly conduct various technical events for students. Motivated by the recent national and international thrust on research, the department has begun to consider research seriously. As a result, a good number of faculty members obtained doctoral degrees, and the remaining are being pursued. The department is giving high importance to publications which in turn gives impetus to research in the department. In addition, the department has been granted access to the Chips to Startup (C2S) program by MeitY, enabling students and faculty to utilize advanced EDA tools such as Synopsys, Cadence, and Ansys for VLSI design, research, and innovation.

DEPARTMENT ASSOCIATIONS



SRKR ENGINEERING COLLEGE(A)

BHIMAVARAM

(Accredited by NAAC-A* and NBA – ECE, CE, ME, EEE, CSE, IT)

ONE WEEK

FACULTY DEVELOPMENT PROGRAMME ON

"EMERGING TRENDS IN VLSI, IOT AND QUANTUM COMPUTING"

(11th MAY to 16th MAY 2026)



ORGANISED BY

SRKREC

INSTITUTION'S INNOVATION COUNCIL

IN ASSOCIATION WITH

DEPARTMENT OF E.C.E &

CENTER OF EXCELLENCE IN VLSI DESIGN

SRKR ENGINEERING COLLEGE (A)

SRKR MARG, BHIMAVARAM, W. G. DIST.

ANDHRA PRADESH—534204

NBA, NAAC Accredited, NIRF RANKED

Permanently Affiliated to JNTUK

Approved by AICTE

www.srkrec.edu.in

Faculty Development Program (Online)

Theme: Emerging Trends in VLSI, IoT and Quantum Computing

Duration: 11–16 May | **Time:** 5:00 PM – 7:00 PM

11 May: Convergence of VLSI, IoT & Quantum Computing – Dr. Ravichand

12 May: Latest Trends in Automotive Semiconductors & Post-Silicon Validation – Dr. Venkata Anuroop Theeda (NXP Semiconductors)

13 May: Low-Power VLSI Design for IoT & Clock Domain Crossing (CDC) – T. Subramanyam (Texas Instruments)

14 May: Hardware Security in VLSI Circuits – Dr. Sangeetha Singh; Nanotechnology for Quantum Computing – Dr. T. Rajini

15 May: AI-Driven Optimization of High-Efficiency Front-End VLSI Workflows – Dr. G. Challa Ram

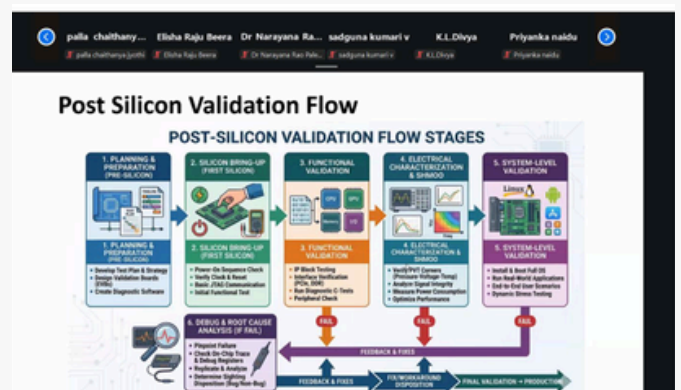
The FDP provided participants with insights into emerging technologies, **semiconductor design, IoT, quantum computing, hardware security, and AI-driven VLSI workflows** through expert sessions by academicians and industry professionals.

DAY-1 INAUGURAL SESSION

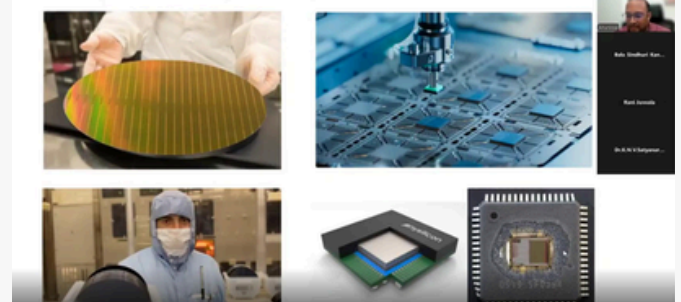


The inaugural session was hosted by **Dr. K. Bala Sindhuri**, who welcomed the participants and introduced **the Chief Guest. Dr. S. Ravi Chand, Professor**, emphasized the importance of emerging technologies such as **VLSI, IoT, and Quantum Computing in shaping intelligent future systems**. The session concluded with a vote of thanks by **Dr. S. Swathi**, expressing gratitude to the speakers and participants.

DAY-2

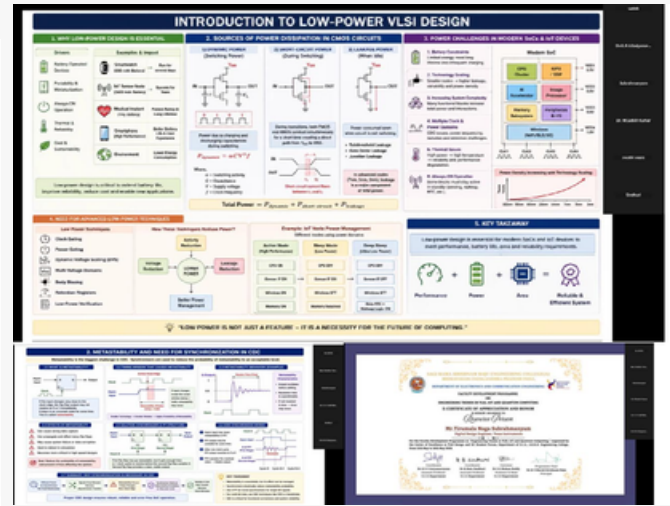


Wafers, Assembly and Others



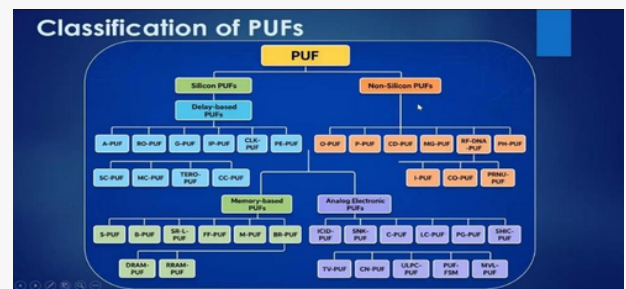
Dr. K. Bala Sindhuri hosted the first session held on **11-05-2026 at 5:40 PM**. During the session, **Dr. S. Ravi Chand, Professor, Department of ECE, D.N.R College of Engineering and Technology**, delivered an insightful talk on the emerging trends in **VLSI, IoT, and Quantum Computing**, highlighting their growing significance in **modern technology and research**. A total of **90 faculty** have attended the **online session**. The session concluded with **Dr. K. N. V. Satyanarayana** expressing heartfelt gratitude to the speaker for providing valuable exposure and knowledge on these **advanced technological domains**.

DAY-3



The **FDP session** was inaugurated by **Dr. K. Bala Sindhuri**, who welcomed the participants and introduced the resource person, **Naga Subrahmanyam Tirumala, Digital Design Engineer, Texas Instruments, Bengaluru**. The online session was attended by **60 faculty members**. The expert delivered a talk on "**Low Power VLSI Design for IoT: Power Domain and Clock Domain Crossing**," highlighting **low-power design techniques, power domain management, and clock domain crossing for reliable and efficient VLSI systems**. The session offered valuable **industry-oriented insights into modern VLSI design**. The programme concluded with remarks by **Dr. K. Bala Sindhuri**, followed by a vote of thanks by **Dr. M. Satish Kumar**.

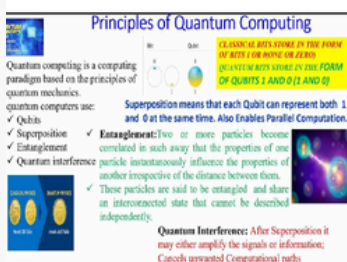
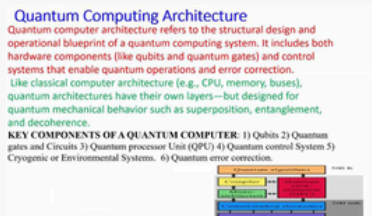
DAY-4



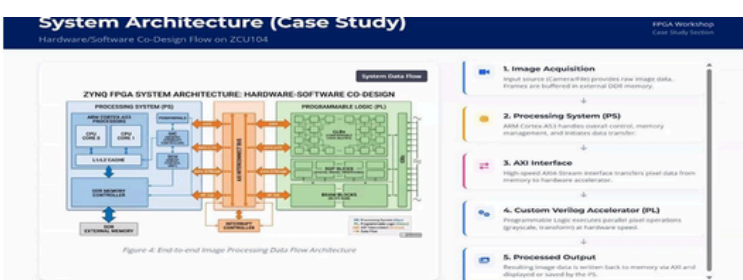
The **FDP session** was inaugurated by **Dr. K. Bala Sindhuri**, who welcomed the participants and introduced the resource person, **Dr. Sangeetha Singh, Assistant Professor, Vardhaman College of Engineering, Hyderabad**. The online session was attended by **60 faculty members**. The expert delivered a talk on "**Hardware Security in VLSI Circuits**," covering **IoT security, classifications of Physical Unclonable Functions (PUFs), and system-level security threats**. The session provided valuable insights into **modern hardware security techniques** and challenges in developing secure **VLSI systems**. The programme concluded with remarks by **Dr. K. Bala Sindhuri**, followed by a vote of thanks by **Mr. T. V. Syamala Raju**.

DAY-4 2ND SESSION

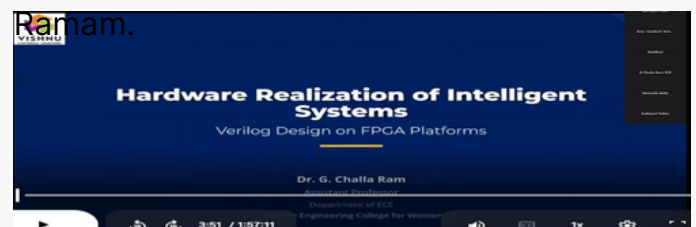
The **FDP session** was hosted by **Dr. K. Bala Sindhuri**, who welcomed the participants and introduced the resource person, **Dr. T. Rajini, Sr. Assistant Professor, VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad**. The online session was attended by **60 faculty members**. The expert delivered a talk on **"Nanotechnology as the Enabler of Quantum Computing: Materials, Devices and Architectures,"** highlighting the role of **nanotechnology, advanced materials, nanoscale devices, and emerging quantum architectures**. The programme concluded with remarks by **Dr. K. Bala Sindhuri**, followed by a vote of thanks by **Mr. Y. Sri Ramam**.



DAY-5 SESSION



The **FDP session** was hosted by **Dr. K. Bala Sindhuri**, who welcomed the participants and introduced the resource person, **Dr. G. Challa Ram, Assistant Professor, Shri Vishnu Engineering College for Women, Bhimavaram**. The online session was attended by **60 faculty members**. The expert delivered a talk on **"Hardware Realization of Intelligent Systems: Verilog Design on FPGA Platforms,"** covering **FPGA-based implementation, Verilog HDL design methodologies, hardware realization techniques, and practical applications**. The programme concluded with remarks by **Dr. K. Bala Sindhuri**, followed by a vote of thanks by **Mr. Y. Sri Ramam**.



DAY-5 2ND SESSION

S.R.K.R ENGINEERING COLLEGE(A), CHINNAMIRAM :: BHIMAVARAM
DEPARTMENT OF E.C.E
CENTER OF EXCELLENCE IN VLSI DESIGN

WELCOME

A ONE WEEK FACULTY DEVELOPMENT PROGRAM
ON
"EMERGING TRENDS IN VLSI, IOT AND QUANTUM COMPUTING"

11TH TO 16TH MAY 2026

Resource Person



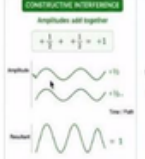
Dr. Appikonda Mohan Durga Kumar
 Associate Professor, Swarnandhra College of Engineering and Technology.

Quantum Interference

Quantum states behave like waves. Their probability amplitudes can add or cancel each other.

CONSTRUCTIVE INTERFERENCE

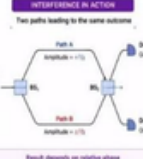
Amplitudes add together

$$+ \frac{1}{2} + \frac{1}{2} = +1$$


Result: Increased probability

INTERFERENCE IN ACTION

Two paths leading to the same outcome

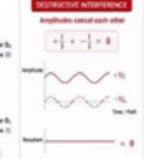


Result depends on relative phase

- If amplitudes add (in phase) → **constructive interference**
- If amplitudes cancel (out of phase) → **destructive interference**

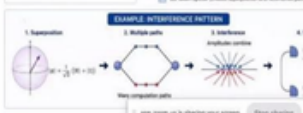
DESTRUCTIVE INTERFERENCE

Amplitudes cancel each other

$$+ \frac{1}{2} - \frac{1}{2} = 0$$


Result: Decreased probability

EXAMPLE INTERFERENCE SET-UP



KEY POINT

Quantum interference is the mechanism that enables quantum computers and enables emerging, shiny quantum algorithms that provide advantage.

The session of the **FDP** was hosted by **Dr. K. Bala Sindhuri**, **Coordinator of the programme**, who welcomed the participants and gave a brief introduction about the **Chief Guest** and **resource person**, **Dr. Appikonda Mohan Durga Kumar**, **Associate Professor, Swarnandhra College of Engineering and Technology**. A total of **60 faculty** have attended the **online session**. The technical session was delivered on the topic **"Introduction to Quantum Computing: Superposition, Entanglement and Quantum Interference."** The speaker explained the fundamental principles of **quantum computing**, including **superposition**, **entanglement**, and **quantum interference**, and highlighted their significance in **next-generation computing technologies**. The session provided participants with valuable insights into the concepts, applications, and future potential of quantum computing. **Dr K.Bala Sindhuri** concluded the session and **Mr. Y. Sri Ramam** has proposed vote of thanks to the chief guest for sharing her valuable insights.

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE(A)
 BHIMAVARAM-534204, ANDHRA PRADESH INDIA,
 DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

FACULTY DEVELOPMENT PROGRAMME
ON
ENGINEERING TRENDS IN VLSI, IOT AND QUANTUM COMPUTING

E-CERTIFICATE OF APPRECIATION AND HONOR
IS HEREBY AWARDED TO
Resource Person
Dr. Appikonda Mohan Durga Kumar
 Associate Professor, Swarnandhra College of Engineering and Technology.

For the Faculty Development Programme on "Engineering Trends in VLSI, IoT and Quantum Computing" organized by the Center of Excellence in VLSI Design and IIC Student Forum, Department of E.C.E., S.R.K.R. Engineering College, from 11th May to 16th May 2026.


Coordinator
Dr. K.N.V. Satyanarayana
 Assistant Professor
 E.C.E Department


Coordinator
Dr. K. Bala Sindhuri
 Associate Professor
 E.C.E Department


Convener
Dr. S.S. Mohan Reddy
 Professor & Head
 E.C.E Department


Programme Chair
Dr. K.V. Murali Krishnam Raju
 Principal

FDP on
Introduction to Quantum Computing:
 Superposition, Entanglement and Quantum Interference

Dr. Appikonda Mohan Durga Kumar
 Principle Investigator,
 C-DAC Bangalore Certified Quantum Universe Developer,
 Associate Professor, Department of CSE
 Swarnandhra College of Engineering and Technology (A)
 Email: appikonda.mohan@gmail.com

Qubit

BIT vs QUBIT

Classical Computing vs Quantum Computing

BIT (Classical)

A bit can be in one of two states

0

State 0

1

State 1

At any given time, a bit is either 0 OR 1



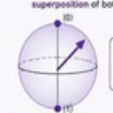
Classical Computer

QUBIT (Quantum)

A qubit can be in 0, 1, or in a superposition of both states

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where $|\alpha|^2 + |\beta|^2 = 1$



At any given time, a qubit can be 0 AND 1 simultaneously



Quantum Computer

Qubits enable quantum computers to process a vast number of possibilities at the same time!

1 app zoom us is sharing your screen. Stop sharing. Help

Fundamentals of quantum computing

By Dr K Basava Raju, Associate Professor, Anurag University, Hyd.



The inaugural session of the **FDP** was hosted by **Dr.K.Bala Sindhuri, Coordinator of the programme**, who welcomed the participants and gave a brief introduction about the **Chief Guest** and **resource person, Dr. K.Basava Raju, Associate professor, Anurag University, Hyderabad**. A total of **50 faculty** have attended the **online session**. The first technical session was delivered on the topic **“Fundamentals of Quantum computing”**. During the session, the speaker has provided an overview of **quantum computing principles such as qubits, superposition, and entanglement**. The speaker explained the differences between classical and quantum computing along with important **quantum algorithms** and **realworld applications**. The session enhanced participant’s understanding of emerging **quantum technologies** and their future impact on computing and research. **Dr.K.Bala Sindhuri** concluded the session and **Dr.M.Satish Kumar** has proposed **vote of thanks** to the chief guest for sharing his valuable insights.

The collage consists of four screenshots from a Zoom meeting. The top-left screenshot shows a presentation slide titled "Applications of Quantum Computers" with categories like Travel & Logistics, Chemistry, Pharmacology, Climate Modeling, Financial Analysis, and Cryptography. The top-right screenshot shows a Zoom chat window with a discussion about quantum computing claims and truths. The bottom-left screenshot shows a presentation slide titled "AI-IoT Integration" with a diagram of Sensory, Action, and Cognitive layers. The bottom-right screenshot shows a presentation slide titled "Core Principle of Entanglement: Linked beyond Second concept of quantum computing is entanglement" with a diagram of entangled particles.

ECE INNOVXPO 2K26

The Department of Electronics and Communication Engineering, SRKR Engineering College, in association with the Institution's Innovation Council (IIC), organized ECE Innovxpo 2K26 as part of the IIC 8.0 Self-Driven Activity under the theme "Redefining the Future of Electronic Design & Innovation." The one-day offline event was held on 27 April 2026 at the ECE Seminar Hall (T-305), providing an engaging platform for students to showcase innovative prototypes as part of the Innovation Showcase: Prototypes thrust area under Validation and Concept Development. Around 270 students actively participated and presented working prototypes across various domains of Electronics and Communication Engineering. The participants demonstrated their innovative ideas, explained design methodologies, and highlighted real-world applications of their projects. The prototypes were evaluated by faculty members based on creativity, technical feasibility, problem-solving approach, and implementation, while the event encouraged students to apply theoretical knowledge to practical challenges and develop innovative solutions.



SRKR ENGINEERING COLLEGE (A)
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IIC 8.0 Self-Driven Activity
ECE InnovXpo 2K26
- Redefining the Future of Electronic Design & Innovation
organized by
Department of Electronics and Communication Engineering

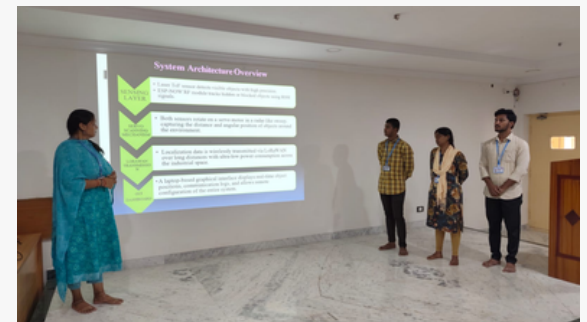
27 Apr, 2026 **Seminar Hall (T-305), ECE Dept.**

Conveners Dr. S.S. Mohan Reddy (Professor & Head, ECE) Dr. K. Tarun Kumar (Convener, IIC)	Faculty Coordinators Dr. B. Sanjay (Assoc. Prof., ECE) Dr. J. Ravi (Member, IIC)	Student Coordinators Miss. V. S. S. Jyothi (Student, IIC) Mr. L. Bhanu Kiran (Student, IIC)
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ECE Innovxpo 2K26 fostered **creativity, teamwork, technical communication, problem-solving, and entrepreneurial thinking** among students. It provided valuable opportunities for **idea validation, peer learning, knowledge sharing, and constructive feedback** from **faculty members**, while enhancing students' **practical and project development skills**. The enthusiastic participation of students made the event highly successful and impactful, reinforcing the department's commitment to nurturing innovation, prototype development, and future-ready engineering talent.



INNOVATION, COMMUNITY, AND SCALABLE STARTUP GROWTH



SRKR ENGINEERING COLLEGE (A)

SRKR MARG, CHINA AMIRAM, BHIMAVARAM, W.G. Dt., A.P., India



IIC 8.0 Self-Driven Activity

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

in association with

INSTITUTION'S INNOVATION COUNCIL

— WELCOME —

to
EXPERT TALK

on
“INNOVATION, COMMUNITY, AND
SCALABLE STARTUP GROWTH”

RESOURCE PERSON



Sriharsha Mallina

Co-Founder : Subha Electronics

Vishesh Country Cache

Visakhapatnam, Andhra Pradesh, India

 **29th June 2026**

Venue:
ECE Seminar Hall,
SRKR Engineering College (A),
Bhimavaram

*Let's Innovate Today for a
Better Tomorrow*

Convener
Dr. S.S. Mohan Reddy
(Prof. & Head, ECE)

Coordinator
Dr. J. Ravi
(Asst. Professor, ECE)

Student Coordinator
V. Skarlet Swarna Jyothi
(IIC Student Convener)



The Department of Electronics and Communication Engineering, in association with the Institution's Innovation Council (IIC), SRKR Engineering College (A), organized an **Expert Talk** on **"Innovation, Community, and Scalable Start-up Growth"** on **29 June 2026** at the **ECE Seminar Hall**. The programme was conducted as part of the **IIC 8.0 Self-Driven Activity** under the thrust area **"Start-up Ecosystem and Scale-up"**, in connection with the Innovation Showcase: Prototypes. The session featured **Sriharsha Mallina, Co-Founder, Subha Electronics, Vishesh Country Cache, and Visakhapatnam**, as the **resource person**. The expert talk focused on the importance of innovation and community engagement in identifying real-world challenges and transforming them into entrepreneurial opportunities. The resource person shared **valuable insights into prototype development, product validation, market analysis, customer-centric design, funding opportunities, intellectual property awareness, incubation support, and scalable start-up models**. Drawing from his entrepreneurial experience, he motivated students to develop innovative solutions with meaningful social and industrial impact.



The programme witnessed enthusiastic participation from **80 students** and **25 faculty members** from various departments, making it a valuable **multidisciplinary learning experience**. **Students actively interacted** with the **resource person** through discussions and a question-and-answer session, gaining practical knowledge about **start-up ecosystems, product innovation, prototype commercialization, and emerging technologies**. They were also encouraged to utilize institutional innovation platforms and participate in prototype exhibitions, hackathons, start-up challenges, and incubation programmes. The event was coordinated by **Dr. J. Ravi, Department of Electronics and Communication Engineering**, under the convenorship of **Dr. S. S. Mohan Reddy, Head, ECE**. The session successfully enhanced awareness of **innovation-driven entrepreneurship, prototype development, and scalable business models**, while encouraging participants to explore entrepreneurship as a promising career path. Overall, the **expert talk** served as an inspiring platform for promoting innovation, interdisciplinary collaboration, and entrepreneurial excellence, motivating students and faculty to transform **creative ideas** into **impactful technological solutions**.